

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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CIVIL WORKS SPECIFICATIONS

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Revision History

Rev No.	Date	Description of Revision
R01	04.07.22	Issued for Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-CIV-SPC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
BS	British Standard
DEP	Design & Engineering Practice
ASTM	American Society for Testing and Materials
AWS	Association of Welding Society
ASCE	American Society of Civil Engineers
PIP	Process Industrial practices

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

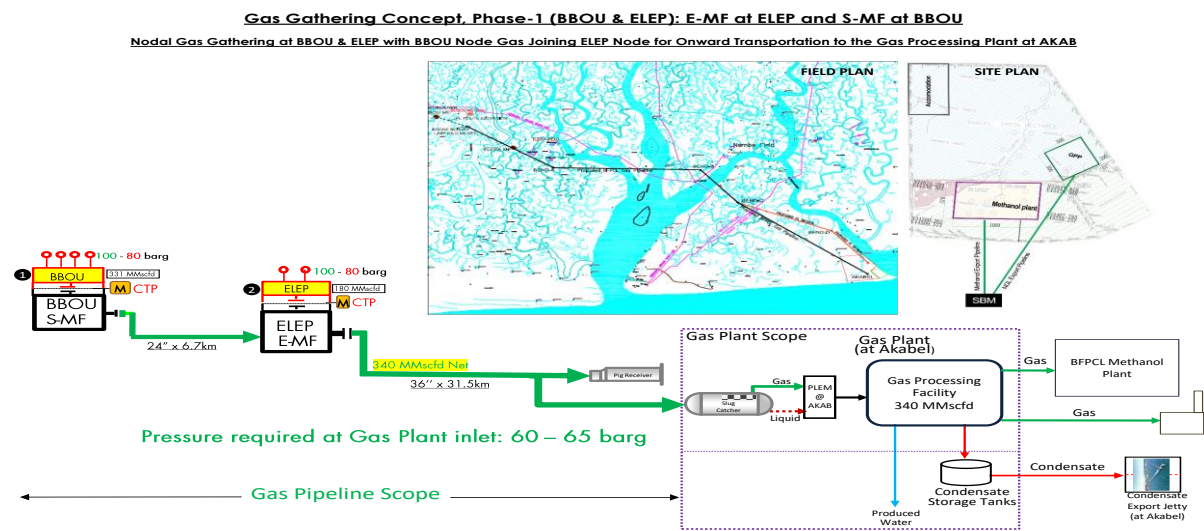


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification is to provide the minimum requirements for the design, construction, installation and completion of the Civil Works for proposed pipe line Project.

This specification shall be read in conjunction with the structural specifications. For the purpose of this specification, the name “Civil works” includes all activities to be done on the soil of the proposed site including the materials to be used.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
References	Title
DEP 34.17.00.32- Gen.	Design and Engineering Of Buildings
DEP 34.28.00.31-Gen.	Onshore Steel Structures
DEP 34.19.20.31-Gen.	Reinforced Concrete Structures
BS EN 1992-1-1	Design of concrete structures – Part 1-1: General rules and rules for buildings
BS EN 1993-1-1	Design of steel structures - Part 1-1: General rules and rules for buildings
BS EN 1997-1	Geotechnical design - Part 1: General rules
BS EN 1991-1	Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings
BS 8666 – 2005.	Steel Reinforcement For Concrete
BS 6349-1:2000	Code of practice for Maritime structures- Part 1: Code of practice for general criteria.
BS EN 934-6:2001	Admixtures for concrete, mortar and grout. Sampling, conformity control and evaluation of conformity
ASTM C150 / C150M - 12	Standard Specification for Portland cement
BS EN 206:2013	Concrete. Specification, performance, production and conformity
BS EN 12620:2002+A1:2008	Aggregates for concrete
BS EN 1008:2002	Mixing water for concrete. Specification for sampling, testing and assessing the suitability of water, including water recovered from processes in the concrete industry, as mixing water for concrete

ASCE 7 -05	Minimum Design Loads for Buildings and Other Structures
BS 6063	Code of practice for earthworks
ASTM A 185/A185M	Standard Specification for Steel Welded Wire Reinforcement for Plain Concrete
BS 1377	Methods for Test for Soils for Civil Engineering Purpose
BS 8110	Structural use of concrete
BS 8010-2	Code of Practice for Pipelines-Pipelines on land: Design, Construction and Installation, section 2.8 Steel for Oil and Gas.
BS 729	Specifications for Hot Dip galvanized Coating on Iron & Steel
ASTM D 1557	Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort
PIP STC 01015	Process Industrial Practices Structural Design Criteria
BS 4449	Specification for Carbon Steel Bars for the Reinforcement of Concrete
BS 4	Structural Steel Sections Part 1 –Specification for Hot Rolled Sections
ASTM D 1143	Methods of testing piles under static axial compressive loads
BS EN 1994	Design of composite concrete and steel structures

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Concrete Works

3.1. Scope

This specification is to provide the minimum requirements for the construction, installation and completion of the Concrete Works for proposed pipeline Project.

3.2. Materials

3.2.1. Cement

Cement for foundations and reinforced concrete structures on land and in contact with seawater shall be ordinary Portland cement according to BS EN 197-1:2011.

3.2.2. Concrete Admixtures

Admixtures shall not be used without the approval of Principal. Generally, admixtures should comply with BS EN 934-6:2001 Concrete Admixtures Concrete Admixtures

3.2.3. Water

All water for concrete shall be fresh, clean and free from all matter liable to reduce the strength and durability of concrete according to BS EN 1008:2002 Water for making concrete. Only water of approved quality shall be used for washing, curing of concrete and similar purposes. The use of wash water (recycling water) is not allowed without approval of Principal.

3.2.4. Aggregates

Aggregates for concrete shall conform to BS EN 12620:2002+A1:2008. Potential alkali reactivity shall be determined with ASTM C 295 and ASTM C 289. Coarse aggregates shall be stored on site on clean platforms free from direct contact with the ground and in separate bins or batches for each size. Prior to use all aggregates shall be washed mechanically with water of mixing water quality to remove clay, silt and fine dust.

3.3. Concrete Mixes and Concrete Grades

3.3.1. Concrete Mixes

All concrete shall comply with the provisions of the latest versions of BS EN 1992-1-1 and ASTM C 150. The grade of concrete works shall be as follows: - Blinding Concrete: Grade 15 ($f_{cu} = 15\text{N/mm}^2$). Reinforced Concrete: Grade 25 ($f_{cu} = 25\text{N/mm}^2$) minimum and Grade 30 ($f_{cu} = 30\text{N/mm}^2$) maximum.

Mechanical means shall be used for all concrete mixing. Hand mixing shall not be allowed even for small works (small works means concrete volume that is less than 0.2m^3). Concrete mixing equipment shall be sufficient for the size of the structures to be constructed and the concreting schedules, taking into account the need to ensure a continuous supply of concrete for each

structure. Generally, a mechanical means shall be used for measuring the concrete materials, and the concrete composition shall be constant and identical with those that proved to be the optimum during the preliminary tests for each class of concrete.

3.4. Form Work

3.4.1. General

Timber or steel may be used for formwork or scaffolds at the Contractor's convenience or according to the material he has available and shall comply with BS 5975:2008+A1:2011.

Formwork shall be able to withstand the weights of the structure, of the workers, and fixed and mobile equipment for the placing and compaction of the concrete, as well as any other loads or stresses. Moreover, formwork shall be erected to prevent distortion or misalignment during concreting, and shall be built to allow the striking of the formwork in accordance with the provisions of BS EN 1992-1-1.

Care shall be taken in the design, erection and dismantling of the formwork and its supports when heavy loads, wide spans, or similar unusual features are involved. The formwork surface in contact with the concrete shall be smooth and free from projections, indentations, undercuts or unevenness, care being taken to accurately mate the boards so that the required finished surfaces may be obtained.

Unless otherwise specified, all edges and corners of exposed concrete shall be chamfered using timber fillets (25mm x 25mm). PVC or fibre cement pipes of either round or square section may be used to provide anchoring pockets for holding-down bolts and similar items.

Their outside surfaces shall be greased before concreting to facilitate their easy removal after the concrete has set. Inserts shall be provided with holes for fixing anchor bars together with bottom plugs, when necessary. The pockets so formed shall be thoroughly cleaned before the anchor is installed; pockets shall be filled with polystyrene foam or similar to prevent any damage to the concrete.

Formwork shall conform to the lines and dimensions on the drawings and shall be substantial and sufficiently tight to prevent leakage of mortar. Shutter accessories to be partially or wholly embedded in the concrete, such as ties and hangers, shall be of a commercially manufactured type.

The suitability of the form release agent shall be verified by site trials considering the quality of the concrete surface, the formwork material and the climatic conditions. Formwork shall be rigidly constructed and be mortar-tight. Timber for formwork shall be well-seasoned and of good quality.

3.4.2. Removal of Formwork

The work of striking (dismantling) the formwork shall only be carried out under the supervision of a competent foreman. The striking periods for cast in place concrete shall be based on BS EN 1992-1-1.

The striking period for the single structures have to be confirmed by design criteria and verified by strength tests on cubes cured under the same conditions as the concrete in the member as far as possible. Props shall remain to decrease deforming. The curing shall take place immediately after 18 hours from casting.

3.5. Reinforcement

All reinforcement shall be grade 460 ($f_y = 460\text{N/mm}^2$) according to BS4449 or higher. The Construction Contractor shall supply Principal with mill order sheets and test certificates. Steel reinforcing bars and mats shall be kept clean and shall be free from pitting, loose rust, mill scale, oil, grease, mortar, earth, paint and any other material which may impair the bond between the concrete and the reinforcement, or which may cause corrosion of the reinforcement or disintegration of the concrete.

No steel shall be stored directly on ground. All bar reinforcement shall be bent cold before the bars are placed in position. No heating or warm bending will be permitted. Only experienced steel benders and fixers shall be employed. Site welding of reinforcement is to be approved by Principal.

Bar reinforcement shall be shaped to the exact dimensions required. All bending tolerances shall comply with the relevant British Standards and regulations or approved equivalent.

3.6. Embedded Items

All anchor holding-down bolts, rag bolts, steel angles and the like shall be supplied and built into the concrete as required by the design.

All embedded parts are to be placed accurately and are to be fixed so that they remain in position during concreting.

All embedded parts have to be corrosion protected according to their exposure condition and the relevant British Standards or codes and regulations. All embedded parts shall be corrosion protected by hot dip galvanizing.

3.7. Concrete Batching Plant

The measuring, mixing, production and transporting of concrete shall be in accordance with relevant codes and standards (BS EN 206:2013, BS EN 1992-1-1).

The weight batching equipment used shall be approved types accurate to within 3% and checked

in consequent intervals specified in the inspection and test plan. Volumetric batching is not allowed. All measuring equipment of the batching plant shall be calibrated prior to start of concrete works and shall be maintained in a clean and serviceable condition.

The weighting devices shall be zeroed daily, checked for accuracy monthly and calibrated bi-monthly. The moisture content of the aggregates shall be determined at the daily start of production and at changing moisture conditions of the aggregates.

The concrete shall be mixed in a batch type or continuous mixer capable of thoroughly combining the aggregates, cement and water into a uniform mass within the specified mixing time and of discharging the concrete without harmful segregation.

The batch plant shall be equipped with an automatic data recorder which shall, as a minimum, record the following information: grade of concrete, date, time, and cumulative production for each grade of concrete, truck number, cement and aggregate scale tare, individual weights of cement, each type of aggregate, admixture, water and the moisture compensator setting.

3.7.1. Transportation of Fresh Concrete

For the transportation of fresh concrete only approved transport vehicles (truck mixer) and containers are permitted.

No truck mixing is allowed. All truck mixers used for transport of concrete, concrete pump hoses and other equipment for concrete transport shall be protected against solar radiation. Delivery tickets showing all relevant information for identifying the concrete mix and the related production record shall be used.

3.7.2. Placing Concrete

Prior to concreting, checks shall be carried out to determine that:

The steel reinforcement fully complies with the construction drawings with regards to the number, location and diameter of the bars, their bending, joints, offsets, laps, spacing, cover and tying, and that the cages are fastened so as to ensure their stability during concreting;

All the formwork has been thoroughly cleaned and all foreign matter has been removed;

All inserts, as indicated on the construction drawings, have been correctly located inside the shuttering; That all holding-down bolts, stays, sleeves, plates, pipes, etc. have been correctly positioned to the proper levels, position and alignment, on both horizontal and vertical structures, above and below ground, as required for the subsequent erection of any permanent or other temporary equipment;

The formwork and any other tiling and cement material is kept damp, especially during hot weather;

All flaws, distortions, cracks etc., caused by board shrinking and/or settling have been properly eliminated from timber formwork, particularly during hot weather;

All appropriate, mobile and stationary equipment has been properly installed for concrete lifting, conveying and distribution, to ensure concreting as scheduled without interruption; avoid shock loading and/or vibration from this equipment which could affect concrete stability and curing;

Where large pours are involved, requiring several days to complete, the limit of the daily pour shall be scheduled so that the construction joint occurs in the most appropriate location, depending on the type of structure.

No concreting shall be commenced without the approval of Principal. The Construction Contractor shall give a 48 hours' notice of time and location of concrete placing.

The rate of delivery of concrete during concreting operations shall be such that the interval between batches shall not exceed 20 minutes. Concrete shall be brought up in layers not exceeding 300 mm in depth. Each layer shall be placed and consolidated before the previous batch has taken initial set to prevent injury to the green concrete and to avoid surfaces of separation between the batches.

The method of compaction has to be selected according to the intended placing method and consistency of the fresh concrete. In general internal vibrators will be used for compaction. In addition, vibrator screeds will be used for a finishing of horizontal surfaces. Skilled operators only shall be employed in vibrating the concrete and they shall be subject to constant supervision. No concrete shall be allowed to fall uncontrolled through a height of more than 2.00m.

Unless adequate protection is provided and approval is obtained, concrete shall not be placed during a heavy rainfall. Rainwater shall not be allowed to increase the mixing water or to damage the surface finish.

3.8. Curing of Top Surfaces

Every part of the exposed surface shall be covered with a layer of polythene or similar impermeable sheeting immediately after finishing. The sheeting shall be in direct contact with the concrete, unless some initial separation is necessary to obtain a particular surface finish. The sheets shall overlap adjacent sheets at the edges for at least 200 mm, and they shall be fixed down at the ends and edges to prevent air circulation over the concrete.

As soon as the concrete has stiffened sufficiently an additional layer of wet Hessian with a top layer of polythene or similar impermeable sheeting will be applied.

3.9. Quality control

The Construction Contractor shall submit at least two months before any concreting a "Quality Control Programme".

The Construction Contractor shall assign an approved engineer by Principal with dedicated responsibilities for all QA/QC aspects of the work.

An approved material laboratory for the purpose of carrying out all required tests has to be provided by the Construction Contractor.

Inspection reports for each individual concrete pour shall be submitted to Principal. The preparation checklist shall comprise details on formwork, reinforcement, concrete cover, cleaning, equipment, staff etc.

The concrete quality report shall contain details of concrete composition from trial mixes and work mixes, location of concrete work, slump test, compressive strength results, weather conditions, finishing, curing etc

3.10. Quality Control Tests

The Construction Contractor shall arrange for carrying out tests on the concrete works in accordance with BS EN 206:2013, and all constituent materials namely: cement, aggregates (coarse and fine), water and reinforcements; and shall supply all materials and equipment necessary for carrying out the tests.

For each batch of concrete (5m^3 - 10m^3), test cubes (150mm x 150mm x 150mm size) shall be made, and shall be crushed as follows: -

Three (3) shall be crushed at seven (7) days

Three (3) shall be crushed at twenty-eight (28) days.

4. Piling

4.1. Scope

This specification is to provide the minimum requirements for the construction, installation and completion of the Piling Works for proposed pipeline Project.

4.2. Setting Out

The Contractor shall provide all labour, pegs, rods, survey instruments, concrete posts etc. needed for setting out the works. The Contractor is to ensure that boundary marks defining the limits of the Employer's property are in their correct positions and shall employ a licensed surveyor to check the accuracy of these positions. The Contractor shall be responsible for setting out the works from the drawings and boundary marks.

The Contractor shall be responsible for safeguarding the position and level of all reference pegs, boundary and benchmarks used for setting out the works.

The Contractor shall obtain the Principal's prior agreement on the locations and numbers of reference baselines and datum levels required for setting out of piles. The Contractor shall engage a licensed surveyor to set out on site the positions and levels of each pile. The Contractor shall obtain the Principal's prior agreement on the locations and numbers of boundary marks.

The system of boundary marks shall be employed for cross-checking in conjunction with the method of setting out from reference baselines for each and every perimeter pile to ensure these works are confined within the limits of the Principal's property. Boundary pegs shall be maintained in good condition at all times. While such pegs may be removed temporarily at localised zones to facilitate construction, the Contractor shall re-install them at the earliest possible time. Boundary pegs shall not be removed without prior consultation with the Principal.

4.3. Underground Services

Prior to commencement of work, the Contractor shall establish the locations and levels of all existing underground services within and surrounding the site likely to be affected by his piling operations. Drawings of buried sewer lines, electricity and telephone cables, and water mains as they are known to the Principal shall be made available for the Contractor's review.

These drawings are indicative only and are intended to represent an approximate guide for the Contractor's own verification on site. The accuracy and completeness of the information shown in these drawings are neither guaranteed nor implied. The Contractor shall be responsible for taking all reasonable and necessary steps to confirm and supplement the information provided to make them complete and accurate. No subsequent claim from the Contractor on the basis of incomplete or inaccurate information shall be accepted.

The Contractor shall exercise reasonable skill and care during piling operations to minimize disturbance of existing underground services. The Contractor shall be responsible for and shall bear the cost of any claims for damages to underground services arising from his piling works.

4.4. Equipment and Labour

The Contractor shall provide all frames, equipment, lifting devices and labour necessary for the installation of piles. The Contractor shall satisfy the Principal regarding the suitability, efficiency and operational capability of his piling equipment. The Contractor shall be required to provide adequate number of operational piling frames to ensure that the works are completed within the time period stipulated in the approved construction programme. The Contractor is deemed to have made provision for the availability of standby plant at all times to allow for contingency in case of equipment failure. Equipment found to have a consistent record of breakdowns shall be removed from the Site.

The Principal shall order the removal or replacement of any equipment or staff whenever he is of the opinion that such equipment or staffs are not suitable for the works. The Contractor shall comply with the Principal's instructions on these matters without extra cost or time to the contract.

4.5. Termination of Piles

The proposed pile embedment depth shall be as specified in the geotechnical Investigation report. Where a suitable stable stratum is not encountered after reaching the proposed pile embedment depth, piles shall be terminated with a minimum embedment of 2.40 m into sand strata having an SPT N-value as per soil investigation report or as per design requirements.

For confirmation of the similar strata, two more SPT shall be conducted thereafter within an interval of 1.0 m. Conducting of SPT shall be done at least once in five (1 in 5) pile locations. In case a medium hard or hard stratum is encountered before the minimum embedment depth stated above, the pile shall be terminated in this layer with a minimum embedment of 500 mm into the medium hard strata or 200 mm in hard strata. The level at which a particular strata is encountered and the termination depth in all cases shall be certified by Principal and shall be binding on the Contractor.

4.6. Materials

All materials which may be used in the Plain or Reinforced Cement Concrete work shall be of standard quality conforming to BS unless otherwise approved by the Principal. The Contractor shall make sure all materials are approved by Principal prior to their procurement and before actual use. The Principal shall have the right to determine whether all or any of the materials offered or delivered for use in the works are acceptable. Any material brought to site not in conformity with specification of Principal shall be rejected and the Contractor shall have to remove the same immediately from site at his own expense.

4.7. Installation

Installation of piles shall be carried out in accordance with the approved pile layout drawings as well as relevant sections of BS EN 1997.

To ensure dense and sound concrete up to cut-off level, concreting shall extend to a minimum of 600 mm above the pile cut-off level. However, no extra payment shall be made for this and quoted rates shall be inclusive of this.

Where concrete piles are to be placed in the dry, holes shall be de-watered in such a manner as to prevent any disturbance to the foundation material or to the concrete when placed. Liners shall be provided and installed by the Contractor for this purpose if needed and shall be advanced to a sufficient depth below the founding level so as to completely seal the holes and prevent any water seepage into the holes after they have been pumped out.

Where tremie concrete are to be placed in bored piles the Contractor shall maintain an adequate head of water within the holes to prevent the inflow of water through the bases or walls until the concrete has set.

Where removal of liners is done, they may be removed or left in place at the Contractors discretion. Liners shall remain the property of the Contractor. When a liner is removed, the placing of concrete during removal shall be so arranged that the head of concrete will be at least equal to the hydrostatic pressure at the tip of the liner and prevent ingress of deleterious material

In cases where the reinforcement cage is made up of more than one segment, the same shall be assembled by good butt welding only, where facilities on site are insufficient to make good butt welding practicable, the joint may be made by lapping. The reinforcement at the head of the pile will need to be exposed for a distance of 40 times the bar diameter and the new bars overlapped for this distance. Vertical reinforcement shall project 50 times its diameter above the cut-off level.

All precautions for obtaining clean and sound pile shaft shall be strictly observed. The Contractor shall provide all the equipment, tools and plants required for the work and rates shall include the cost of bringing the equipment etc., to site, proper maintenance and removal of the same after completion of work. All the excavated materials including debris, loose earth, muck etc. shall be carted away and disposed by the Contractor out of the premises to any suitable place as per local statutory authorities.

4.8. Pile Testing

4.8.1. Pile Integrity Test

The integrity test shall be carried out on concrete piles after installation. Either by impulses or vibrations applied to the pile and measurements made of the timing and attenuation of reflected signals. The methods should provide an indication of the soundness of the concrete but they should

be undertaken by persons experienced in the method and capable of interpreting the results with specific regard to piling

The load testing apparatus for applying axial loads to piles shall be capable of maintaining constant load under increasing settlement and of applying loads such that lateral forces or impact will not occur. The measurement of the load shall be by load cell(s) or equivalent device(s) such that temperature effects are eliminated. The readout of the load cell or equivalent device shall be accurate to within 450 N. Determination of absolute or relative pile movement shall be by dial indicators (gauges) which read to .025 mm. The accuracy of the measurement system is to be within .025 mm. Calibration reports shall be furnished for all testing equipment for which calibration is required. When loads are applied with hydraulic jack(s), the jacking system including the hydraulic ram(s), couplings, hydraulic pump, and pressure gauge shall be calibrated prior to the test so that the load applied is controllable to within 5 per cent of the total applied load.

4.8.2. Initial Load Test

Initial Load test may be used for confirmatory test for the Geotechnical Investigation Report. The load can be carried to failure. A loading test carried to failure gives the ultimate bearing capacity of the particular pile that has been tested. To infer the ultimate bearing capacities of other piles on the same site from loading tests alone requires either the prior testing of an adequate number of piles and the use of a site investigation that is sufficiently detailed to show the uniformity or otherwise of the soil, thus enabling the ultimate bearing capacities of other piles to be estimated.

4.8.3. Routine Load Test

Routine load tests shall be carried out up to one and half times the design load. The piles for the test shall be randomly selected by the Principal.

4.8.4. Pile Load Test Details

The Contractor shall issue a method statement for pile load test and this shall be approved by the Principal before any test is carried out.

- i) For pile load tests, the test set up shall be as per BS EN 1997-1:2004.
- ii) All testing shall be done by direct method of loading in successive increments as per the relevant clauses of BS EN 1997-1:2004.
- iii) Each stage of loading shall be maintained till the rate of movement is less than 0.2 mm per hour.

- iv) Test shall be carried out at the cut-off level

4.8.5. Number of Tests Required

The number of tests required shall be as per relevant clause BS EN 1997-1:2004.

A full record of pile load test results shall be submitted to the Principal immediately on completion of each test. The record shall also include the plots of load settlement (for various stages of loads) characteristics of pile and also the interpretation of the pile load test curve as per criteria for safe loads mentioned in the. Any special observations shall be duly explained by the contractor.

4.9. Stripping or Cutting Of Piles to Cut-Off Levels

Before casting the pile cap, the excess concrete shall be cut off up to pile cut-off elevation. Piles shall be cut off at level true to elevation shown or specified on the drawings. Care shall be taken not to damage the reinforcement or the concrete below cut off elevation during such stripping operations. Where stripping is to be done to a level lower than the specified cut-off elevation to obtain dense and sound concrete, the Piling Contractor shall built up the pile up to cut off elevation at his own cost. Stripping of concrete piles to cut-off level specified in the drawings shall be carried out by the Contractor.

Unless otherwise directed, cut-off level shall be 100 mm above the underside of pile cap and all cut-off piles shall be disposed as waste material.

The contractor shall cut the pile head with mechanical cutter to form flush cut surface and construct the starter bars bonded to the annulus of the hollow section precast spun pile as per Drawings. In the Principal's opinion, if the pile is defective, the Contractor shall carry out his own expenses on all necessary remedial work as required by the Principal.

Upon completion of the piling work, all casing equipment, construction tools, protective covering and debris resulting from the piling operations shall be removed from the works site with the permission of the Principal.

4.10. Measurement of Piles

Piles shall be measured and paid for based on the actual pile length from pile tip to the cut off level as per the approved working drawings with the contractor according to approved design or as indicated by the Principal. No extra payment shall be made for empty boring, if any.

For the purpose of measurements, the length of pile shall be measured from the tip of the pile to the bottom of the pile cap. The 100 mm length of the pile projecting into the cap shall not be measured and also shall not be deducted from the quantity of pile cap concrete. Payment for

routine load tests which shows unsatisfactory results shall not be made.

Piles showing unsatisfactory results shall be treated as defective piles. Defective piles shall be removed or left in place and replaced by additional piles as directed by Principal at no additional cost to the owner. Reaction piles, if required for the purpose of conducting load tests on piles shall not be paid extra. Quoted rate for tests shall be inclusive of this. However, for routine lateral and uplift tests, if required, job piles may be used as reaction piles.

4.11. Recording of Data

Daily site records shall be maintained by the Contractor for the installation of piles against each rig and shall contain the number and dimension of the pile, depth bored, time taken for boring, concreting and empty boring (cut-off level), chiseling and where the pile is wet or dry.

Typical data sheet in triplicate for recording piling data are to be prepared by the Contractor before starting of actual piling operation at site and information to be recorded against each pile in the data sheet shall be signed by the Contractor and countersigned by Principal. One copy of the data sheet shall be retained by the Contractor and the other two copies to be submitted to Principal for records and future reference.

At least six photographs are to be taken by the party depicting progress of the job in every week and a set of same shall be separately submitted for review and record to the Principal.

Individual pile records shall be submitted not later than noon of the next working day after the pile was installed. The signed records shall form a record of the work. Any unexpected installation conditions shall be noted in the record.

The data to be recorded shall include:

- The dimensions of the piles, including the reinforcement detail and the mark and location of the pile
- The boring method employed
- The type of soil in which pile is terminated.
- The depth bored
- The depth of water table
- The time taken for concreting
- The cut-off level/working level
- The consumption of cement.
- Pile Type (Compression or Tension)
- Date and Time of Installation
- Date of Concreting
- Concrete Mix
- Method of Concreting

- Standing Ground water Level
- Weather Condition
- Ground Level before Commencement of Pile Installation
- Working Level of Ground for Pile Installation
- Depth from Working Level to Pile Toe
- Level of Pile Toe
- Pile Cut-off Level
- Depth from Cut-off Level to Pile Toe (Pay Length)
- Length of Temporary Casing
- Length of Permanent Casing
- Details of Soil Strata Penetrated
- Soil Samples taken and In-situ Tests carried out
- Length and Details of Reinforcement
- Estimated Volume of Concrete required forming the pile shaft Actual Volume of Concrete used in forming the pile shaft
- Type and Model of Plant, Equipment all records shall bear the names of person who records and person who checks.
- Any other important observations.